Horizon — the full pipeline (everything that was built)

green = math · lavender = LLM / spotlighting · rose = direct / jailbreak layer · white = code · beige = data / software

1 · Data (data/README.md = data card)
Indirect injection data
BIPIA (email · table · code, 15 attack kinds)
LLMail-Inject phase 2 (3,165)
AgentDojo v1.2.1
suites slack · workspace · travel
simulated tools, no real side effects
Corpus + canaries
attacks wrapped in carrier templates
CANARY-xxxxxxxx from secrets (CSPRNG)
Direct / jailbreak data
HackAPrompt (601,757 · 77,936 successful)
HarmBench · JBB · deepset · Gandalf
Tensor Trust · XSTest · Alpaca (benign)
Fixed splits
fit → choose τ^*, π^*, λ · calib · report
never report on the split used to choose
fit / report docs
2 · Stage 1 — keyed spotlighting (offline, chooses π^*)
8 transforms τ
none · delimit · ^ datamark · N^{κ} dynamic marker
ROT13 · shift (25) · scramble $|\Sigma|!$ · Base64
Fresh key κ per prompt
keys.py → secrets
Spotlit prompt
$x_\kappa(\delta) = S_\kappa \,\|\, T \,\|\, \tau_\kappa(D \oplus \delta)$
Attack indicator
$A(\delta,\kappa) = 1[\text{canary appears in output}]$
Utility
U_τ = SQuAD token-F1 on clean docs
Clopper–Pearson
$\bar{p} = \mathrm{Beta}^{-1}(1-\alpha;\, s+1,\, n-s)$
Forging bound
$q / |K_\tau|$ (log-space)
Adaptive attacker (PAIR)
max over $\delta \in \Delta$, $|\Delta| \le$ attack_budget → $\bar{p}^{\mathrm{adp}}$
χ^2-DRO
$g_\rho(p) = p + \sqrt{(2\rho\, p(1-p))}$ (else 1)
Combined risk
$\bar{R}_\tau(\rho) = \min\{1,\, q/|K_\tau| + \max(g_\rho(\bar{p}),\, \bar{p}^{\mathrm{adp}})\}$
Sion game, level 1
$V_1 = \min_\pi \max_{\mathrm{attack}}$ (SOCP, cvxpy/Clarabel) → π^*
Choice rule
$\tau^* = \operatorname{argmin} \bar{R}_\tau$ s.t. $U_\tau \ge U_{\mathrm{none}} - \eta$
Significance
McNemar (paired) · bootstrap CI
Hand-off to Stage 2
π^* · $|K| = \min |K_\tau|$ · $\bar{R}_Q = \Sigma\, \pi^*_\tau\, \bar{R}_\tau$
results/stage1/pi_star.json
π^*, $|K|$
$\bar{R}Q$
0 · Software & infrastructure
uv + Python 3.11
pinned uv.lock · scripts/setup.sh
colibri engine
Qwen3.6-35B-A3B (i4)
writes the report
Math libraries
cvxpy + Clarabel (SOCP) · scipy (Beta^{-1}, linprog)
statsmodels · numpy · Z3
Ollama → Qwen3-8B
RTX 4060 · 16k ctx · q8 KV · flash attn
P-LLM, Q-LLM, attacker
MLflow
results/mlflow.db
Role proxy
developer→system · thinking off
LiteLLM
one client · on-disk cache
6 · Evaluation (§11 runners, logged to MLflow)
Unit tests (§9)
pytest · slide values only in tests/
Smoke test
does every layer fire?
CaMeL baseline
reference code on AgentDojo
stage1_runner
static · clean · adaptive · analyze
agentdojo_runner
ablations none · stage1 · stage2 · stage1+2 · full
calibrate λ · adaptive attacks
jailbreak_runner
detection rate / FPR on every direct set
Metrics
ASR = s/n with CP interval · utility · FPR
PyRIT
direct: PAIR · TAP · Crescendo (judge ≥ threshold)
indirect: XPIA plant δ → Horizon → canary
garak
direct: dan · promptinject · encoding
indirect: latentinjection · web_injection
bare model vs guarded_server.py
7 · Outputs & how you use it
./horizon.sh
menu · Stage 1 / Stage 2 / jailbreak demos
help · glossary · session logs
./run_all.sh
quick · pilot · full · --report-only
Figures F1–F18
matplotlib · PDF + PNG + CSV · formula box
Pipeline diagrams
Graphviz → docs/
Horizon.ipynb
every step, executed (nbclient)
./run_campaign.sh
≥ 5,000 tests · resumable · SLURM job (cluster/)
→ report/campaign/: process + attacks through / blocked
report/report.pdf
colibri model analyses figures + CSVs
numbers checked → LaTeX (xelatex)
3 · Direct injection / jailbreak layer (off-the-shelf, no training)
Output guard
Qwen3Guard-Gen 0.6B · Safe / Controversial / Unsafe
User prompt u
trusted channel · role owner / member / guest
Canary check
secret token leaked ⇒ block
Input guard
DeBERTa-v3 injection classifier (ProtectAI)
XPIA plants δ
Answer to the user
Role rules
policy.yaml: what each role may call
4 · Stage 2 — math-hardened CaMeL (online, every request)
Tool outputs
emails · files · web (untrusted)
Privileged LLM
sees ONLY u → writes a Python-like plan
CaMeL interpreter (STRICT)
google-research/camel, vendored unmodified
tracks data + control dependencies
untrusted x
Quarantined LLM × m
run i: own $\tau_i \sim \pi^*$, own fresh κ_i · no tools
Typed parser
Pydantic schema · strict / loose string caps
QIF leakage
$\Sigma \log_2|S|$ bits ≤ leakage_cap_bits
Majority vote
> m/2 identical → value, else refuse
Agreement score
$s(x)$ = (# agreeing runs) / m
Hoeffding
$P(\text{vote fooled}) \le \exp(-2m(\tfrac{1}{2} - \bar{R}_Q)^2)$
Capability labels
SYSTEM ⊑ USER ⊑ UNTRUSTED
$\mathrm{cap}(f(v_{1..k})) = \sqcup\, \mathrm{cap}(v_i)$
Policy table
config/policy.yaml · strict / confirm_untrusted
Noninterference (Z3)
∀ args: UNTRUSTED ⇒ ¬ALLOW · proven at startup
Conformal risk control
ask iff $s(x) \ge \hat{\lambda}$, else DENY
$\hat{\lambda} = \min \lambda : (n/(n+1))\, \hat{R}(\lambda) + 1/(n+1) \le \alpha_{\mathrm{user}}$
DENY
ALLOW
Ask the user
simulated approve / you in the shell
approved
Tool call
AgentDojo environment
G, C, F per attack
G, C, F per attack
G, C, F per attack
cap
5 · End-to-end Horizon risk
Per-attack loss
$\ell = G + a \cdot C + h \cdot F$
G goal hit · C confirm asked · F false refusal
Loss bound
$\bar{\ell}$ = Clopper–Pearson upper on mean loss
Sion game, level 2 (LP)
$\min_{\pi\in\Pi} \max_w \Sigma\, w\, \bar{\ell}$ · scipy linprog + duals → π^*, w^*
Π = configs with leakage ≤ cap and $U \ge U_{\min}$
Horizon value
$V_H = \min\{1,\, \Sigma\pi\, q/|K| + \max(E[L] + \sqrt{(2\rho\, \mathrm{Var}\, L)},\, \Sigma\pi\, \bar{\ell}^{\mathrm{adp}})\}$
(variance form pending author confirmation)